

Universal BFG Reclosure and the Conditions for Irreducible Emergence

A formal synthesis, an exact novelty theorem, an identifiability result, and a generative test architecture

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Terminology. In this manuscript, BFG means BalanceFeld Gleichung. Older source titles retaining “Balance–Field Framework” are cited under their original names.

Scientific status. This is a theoretical preprint. Exact statements are restricted to their hypotheses; exploratory synthetic calculations are not empirical confirmation of a universal natural law.

Abstract

We connect the generator-level BFG grammar of formation, bounded non-vanishing persistence, and relational reclosure to its finite five-object successor representation. A neutral-compatible stratum admits an exact inheritance–novelty identity: commutation of inherited formation with neutral load is equivalent to unchanged formation, while noncommutation produces strict loss of the second spectral moment. This witnesses internal structural novelty, not automatically an irreducible higher closure. We formalize the latter by source continuity, recursive higher closure, generative removal, and specified mechanistic rivals. A representation theorem shows that invertible recoding of a hidden mediator gives identical observed paths. An earlier constructed pilot has a positive removal effect but is outperformed by a direct dyad. We additionally construct a two-stage synthetic example with persistent sources and successful edge removal and restoration. A matched-memory direct source-history model reproduces its complete trajectories and intervention signatures to numerical precision. Its operational synthetic gate passes, while specific mediation exclusion does not. Universal existence, canonical-map realization, and natural-carrier irreducibility remain unproved.

A Hessian-tangent bridge reconciles the quartic Level-0 formation with exact quadratic neutral elimination: the latter is exact in the explicitly normalized second-order sector, whereas finite-displacement cubic and quartic terms remain in the parent formation energy. This resolves their mathematical compatibility without claiming that Level 0 alone uniquely chooses the neutral-sector rule.

Keywords: BalanceFeld Gleichung; neutral mediation; recursive closure; finite reclosure; structural novelty; causal removal; identifiability; emergence.

1. Introduction and claim discipline

The BFG research line contains two descriptions of closure. The generator-level account states the roles a formation must retain and reapply to relations among closed sources [1]. The finite account constructs a successor from a typed state and may terminate when an exact admissibility gate fails [2]. This paper gives a common logical reading of the two descriptions and separates a generated successor, a structural novelty witness, and an irreducible higher closure. The distinction is needed because the finite theorem concerns the spectrum of an inherited formation operator, whereas the strong emergence claim concerns a new relation among still-active source closures.

The present synthesis introduces no new numerical constant into the BFG finite map. Its removal tests and rival models are criteria for a scientific claim about a particular carrier; they do not constitute an additional BFG generator. Standard linear algebra is used as representation mathematics, not as an external physical premise [11].

Table 1. Claim classes and scope.

Class	What is established	Boundary
Definition	Specifies formation, closure, or emergence predicates.	Does not assert that a carrier satisfies them.
Exact finite theorem	Proves an identity on its stated BFG stratum.	Does not by itself prove a higher source relation.
Synthetic tests	Compare declared constructed generators and controls.	Do not establish natural or canonical-map realization.
Carrier claim	Requires an independent measurement and intervention.	Not established by the current toy models.

2. Generator-level closure and relational reapplication

Let $D(X)$ express constitutive differentiation, $Coh(X)$ constitutive coherence, $N(X)$ neutral compatibility of their coupling, and $\Gamma(X)$ regulated exclusion of non-integrable excess without erasure of the constitutive roles. We use the source predicates from the pure BFG emergence account [1]:

$$F_BFG(X) := D(X) \wedge Coh(X) \wedge N(X) \wedge \Gamma(X). \quad (1)$$

Persistence requires admissible continuations to preserve a nonzero constitutive signature with bounded responses. Writing this property as $Pers(X)$ gives

$$Cl_BFG(X) := F_BFG(X) \wedge Pers(X). \quad (2)$$

A closed source can become a pole of a relation among several closed sources. The same predicates in (1)–(2) are then applied to the relational candidate rather than assuming in advance that every source pair generates a new level. These equations are grammar-level definitions. They specify what a formation and a recursive closure would be, while a carrier-specific generator must produce the relevant trajectory and relation.

For a source family $S=\{C_1,...,C_m\}$, let R_+ denote a proposed constitutive relation and C_+ the candidate higher closure. $Continuity(S,C_+)$ requires each selected source to remain independently closure-effective in the higher configuration. A one-time correlation or an aggregate readout is insufficient for recursive closure [1,7].

3. The finite reclosure representation

The current finite BFG candidate is represented by five typed objects: a Hermitian formation geometry K , formation density ρ_F , normalized identity witness ρ_W , positive neutral load Y , and persistent projector P [2]. Its successor operation is deterministic on its declared admissible domain and returns an absorbing terminal outcome otherwise:

$$\Xi := (K, \rho_F, \rho_W, Y, P), \quad U_BFG(\Xi) \in \{\Xi_+, \perp\}. \quad (3)$$

The canonical neutral pair is complementary:

$$C_N(Y) = (I+Y)^{-1}, \quad B_N(Y) = Y(I+Y)^{-1}, \quad C_N(Y)+B_N(Y)=I. \quad (4)$$

The selected persistent source, reciprocal branch loads, and actual analysis map A determine the active target carrier. The canonical quadratic neutral completion of the later BFG corpus has the following variational form [3,4]:

$$J_A(D, \nu) = \frac{1}{2} \|D - A^\dagger \nu\|^2 + \frac{1}{2} \|\nu\|^2. \quad (5)$$

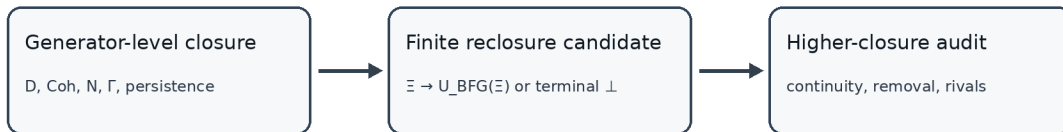
$$\min_\nu J_A(D, \nu) = \frac{1}{2} \langle D, (I + A^\dagger A)^{-1} D \rangle; \quad Y_A = A^\dagger A. \quad (6)$$

Equation (6) follows by solving $(I + AA^\dagger)\nu = AD$ and substituting the minimizer. Y_A is the source-side Gram load. In a declared positive target metric H_+ , the same source-side load grammar reads $Y_A = A^\dagger H_+ A$. On the active carrier, polar identification transports its positive singular spectrum to the target-side successor representation Σ^2 ; the source and target operators are not silently interchanged. The finite successor uses this Gram geometry together with inherited formation transport, identity continuity and a constrained endogenous seed [2,3]. Equation (5) is an explicit canonical quadratic completion; the historical qualitative Level-0 statements alone do not uniquely select its exact form [4,5].

The source places quartic formation in Section 3.1 and quadratic neutral completion in Section 3.2 [13]. Appendix C connects them through the formed branch's Hessian tangent sector and whitened coordinates. Finite-displacement quartic terms remain; using the tangent sector as neutral mediation is an explicit role identification.

$$\dim \mathcal{H}_+ = \text{rank } A \leq \dim \mathcal{H}. \quad (7)$$

The non-increasing active carrier dimension in (7) refers to this finite map. Structural reorganization within an active carrier is logically different from unbounded creation of new basis directions. The abstract generator can classify relations of closed sources, but mapping a particular physical higher-level carrier into the finite state remains a separate representation obligation.



An admissible successor may be inherited, spectrally novel, or strongly emergent.
Each stronger label requires its own stated assumptions and evidence.

Figure 1. Three distinct operations in the present synthesis. The final audit classifies a generated candidate; it does not add a force to the finite successor map.

4. Exact internal novelty and a counterexample to automatic emergence

On the neutral-compatible persistent stratum, diagonalize the restricted load as $Y_P = \text{diag}(y_i)$. For $\alpha, \beta > 0$, the finite reclosure of formation has a Schur-factor representation [2]:

$$(K_+)_{ij} = \chi_{ij}(K_P)_{ij}, \quad \chi_{ij} = (\alpha + \beta y_i y_j) / \sqrt{[(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)]}. \quad (8)$$

$$1 - \chi_{ij}^2 = \alpha \beta (y_i - y_j)^2 / [(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)]. \quad (9)$$

For Hermitian K_P , the second spectral moment is the sum of squared matrix-entry magnitudes in this basis. Consequently:

$$\text{tr}(K_P^2) - \text{tr}(K_+^2) = \sum_{ij} \alpha \beta (y_i - y_j)^2 |(K_P)_{ij}|^2 / [(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)]. \quad (10)$$

Theorem 1 (inheritance and internal novelty). Under the hypotheses of (8)–(10), $[Y_P, K_P] = 0$ if and only if $K_+ = K_P$. If $[Y_P, K_P] \neq 0$, then $\text{tr}(K_+^2) < \text{tr}(K_P^2)$, so the formation eigenvalue multisets differ.

Proof. Each summand in (10) is nonnegative. It vanishes precisely when $(K_P)_{ij} = 0$ for every pair of distinct neutral-load eigenvalues. Those and only those entries contribute to the commutator $[Y_P, K_P]$. In the zero case, $\chi_{ij} = 1$ on all nonzero entries and $K_+ = K_P$. Otherwise at least one summand is strictly positive. For Hermitian matrices, unequal traces of the square imply unequal spectra. $\square$

The theorem supplies a rigorous internal novelty witness. It neither identifies independent source closures nor proves that a new higher relation is causally necessary. The simple case $Y_P = \lambda I$ gives $\chi_{ij} = 1$ for all i, j and therefore $K_+ = K_P$ in this formation step. Thus even strict internal novelty is not automatic for every admissible step. Repeated application with a frozen neutral geometry tends to attenuate cross-eigenspace terms; the complete BFG orbit must reconstruct its state data if it is to exhibit further novelty [2].

5. A conditional criterion for irreducible higher closure

Fix a source family S , a candidate C_+ , a constitutive relation R_+ , a prespecified decision rule Q , and a prespecified class N^* of mechanistically different rivals. The strong BFG classification proposed in [1], with the causal removal requirement made explicit, is

$$E_BFG(C_+|S; N^*, Q) := \text{Continuity}(S, C_+) \wedge \text{Cl_BFG}(C_+) \wedge \text{Need}(R_+; Q|S) \wedge \text{Excl}(N^*; Q). \quad (11)$$

$\text{Need}(R_+; Q|S)$ requires a removal intervention on the generating relation under matched exogenous conditions that reduces the independently defined higher closure while leaving the sources closure-effective. $\text{Excl}(N^*; Q)$ requires a successful comparison with the defined rival class under the same admissible observations and interventions. The classification is conditional on its null class, decision rule, and intervention set; it is not a theorem that every finite BFG successor satisfies (11).

$$U_BFG(\Xi) = \Xi_+ \neq \perp \not\Rightarrow E_BFG(C_+|S; N^*, Q). \quad (12)$$

A valid successor may be inherited, spectrally novel, or a candidate for irreducible emergence. Strong emergence also requires a separately operationalized relation of still-active sources. Conversely, the abstract generator does not assert that its finite five-object representation is the only possible realization of every carrier.

6. Generative removal and source continuity

For a chosen carrier, write source, mediator, and independent higher-channel dynamics schematically as follows, with u_t denoting measured or controlled exogenous input [8]:

$$s_{t+1} = F(s_t, m_t, u_t; \gamma), \quad m_{t+1} = G(m_t, s_t, u_t; \gamma), \quad h_{t+1} = H(h_t, s_t, m_t, u_t). \quad (13)$$

The removal arm sets the localized coupling γ that forms R_+ to zero in the generator and reruns the entire trajectory from matched initial states with matched exogenous innovations. It must not merely set a reported witness h_t to zero after the fact. Define the prespecified higher recovery score Q so that larger values indicate better closure:

$$\Delta_{\text{gen}} := Q_{\text{full}} - Q_{\text{remove}}. \quad (14)$$

A favorable sign in (14) is interpretable as isolated relation necessity only if source persistence meets its separately prespecified gate in both arms. Reintroducing the same relation at fixed parameters should restore the higher endpoint. A separately measured higher state, rather than a product of source readouts or the removed mediator itself, avoids making the result true by definition. Removal of feedback and export pathways can further distinguish recursive closure from transient coupling, but those are additional targeted interventions [8].

7. Observational equivalence and the clone impossibility result

The distinction between a physical relation and a coordinate description becomes decisive when a hidden mediator is used. Consider a system with mediator m , observed sources s , exogenous inputs u , and observed output y :

$$m_{t+1} = G(m_t, s_t, u_t), \quad y_t = H(m_t, s_t). \quad (15)$$

Let T be an invertible transformation of the hidden mediator state and put $z = T(m)$. The transformed system is

$$z_{t+1} = T[G(T^{-1}(z_t), s_t, u_t)], \quad y_t = H(T^{-1}(z_t), s_t). \quad (16)$$

Theorem 2 (observational clone impossibility). The systems in (15) and (16) have identical observed trajectories for corresponding initial states and the same exogenous inputs. Every statistic based solely on those observed trajectories has the same value.

Proof. Induction on t gives $z_t = T(m_t)$. The output expressions then coincide at each step. Applying any deterministic statistic to equal observed paths gives equal values; the same statement holds for their induced distributions under matched random inputs. Transforming an intervention together with T preserves equivalence. $\square$

A rule demanding strictly better predictive or recovery performance than an exact clone is therefore impossible to satisfy, even for a correctly specified full model. An exact clone tests identifiability of a description and should be reported as a representation-equivalence control, outside the class of rivals that must be beaten. Selective measurement and intervention on a physical mediator, plus explicit exclusions of alternative causal paths, are needed for a claim about its specific physical role. Even then, necessity is relative to the admissible interventions and alternatives.

8. Memory-preserving elimination in the exploratory pilot

The new exploratory synthetic pilot [9] used a mediator with difference $d_t = x_{2,t} - x_{1,t}$, $a = 0.7$ and $b = 0.2$:

$$m_{t+1} = a m_t + b d_t. \quad (17)$$

$$m_t = a^t m_0 + b \sum_{k=0}^{t-1} a^{t-1-k} d_k. \quad (18)$$

Substituting (18) into the source equations constructs a direct coupling of source histories with the same memory and exactly the same source and reporter trajectories. This is a change of representation, not proof that a memoryless direct dyad of comparable complexity is equally effective. It does show why output alone

does not uniquely assign a BFG-specific mediator role. The pilot's memoryless direct dyad is a distinct empirical control and happened to perform better on its recovery endpoint.

The reporter in the pilot had its own update equation and noise, but its input was a constructed function of the two source states. It is an independently evolved synthetic candidate, not an independently established irreducible higher closure. Nor was this pilot a numerical execution of the complete canonical five-object BFG operator.

9. Prespecified rivals and a corrected decision rule

Table 2. Rival classes and what each comparison can establish.

ID	Control	Interpretive role
N1	Common driver without proposed relation	Tests shared-excitation explanations.
N2	Additive source histories	Tests whether separate contributions suffice.
N3	Direct dyadic coupling with comparable memory budget	Tests whether the proposed specific mediation is needed.
N4	Circular shift of one source series	Sensitivity test preserving values and circular autocorrelation.
N5	Matched-capacity latent dynamics	Tests added capacity; exact clones are marked separately.

N1–N3 and mechanistically different instances of N5 are rival explanations. N4 is a temporal sensitivity control, not a generative removal intervention. An exact N5 clone is a representation-control tie by Theorem 2, not a null whose failure can be demanded. Model memory, training data, held-out units, accessible inputs, intervention set, and error tolerance must be frozen before confirmatory data are inspected [8,10].

The decision sequence is: (i) full source continuity and recursive higher recovery; (ii) generative removal lowers the independent higher endpoint with source continuity retained; (iii) restoration recovers the endpoint; (iv) mechanistically different rivals fail to reproduce the same observed and intervention signatures within the prespecified tolerance. For repeated independent units, the prospective protocol specifies a one-sided 95% lower confidence bound for the paired removal effect above zero and Holm correction for the comparisons actually tested [8]. A tie with a true clone is recorded without rejecting a potentially physical mediator.

10. Existing synthetic evidence and its limits

The earlier emergence preprint [1] reported a common-driver toy with source readouts $s_i = \text{clip}(0.25 + 0.7z + \eta_i, 0, 1)$ and a higher witness $h = s_1 s_2$. Its aligned criterion was met in 3,984 of 4,000 runs; a structure-preserving circular-shift control met it in 3,344 of 4,000 (83.6%). The common driver remains an explanation of aligned source variation. Setting the already-computed h to zero does not remove an edge in the source generator.

The later exploratory pilot [9] reran the constructed dynamics after removing its mediator and compared 1,000 paired realizations, each 240 steps long, with a perturbation at step 120. The reporter deviation over 30 post-perturbation steps was smaller when recovery was better. Table 3 reproduces its frozen summary values; all arms passed the pilot's source-spread gate.

Table 3. Exploratory reporter recovery in the constructed pilot; smaller is better.

Model	Mean deviation	Interpretation
Full mediator	0.11793	Baseline constructed mediation.
No mediator / common driver	0.12929	Worse than full by 0.01137.
Direct dyad	0.06058	Better than full by 0.05735.
Exact latent clone	0.11793	Identical paths by construction.

The paired no-mediator minus full contrast was +0.01137 with exploratory bootstrap interval [+0.00895,+0.01358]. The paired direct-dyad minus full contrast was -0.05735 with interval [-0.05868,-0.05608]. On 400 held-out realizations, next-reporter RMSE was 0.015774 for full mediator plus relation and 0.015773 for the direct relational predictor. The exact clone's maximum path difference was zero. These intervals are exploratory resampling descriptions, not prospectively registered confirmation.

The pilot supports only the narrow proposition that the constructed mediator has a causal effect relative to its complete removal in that constructed system. The better direct dyad prevents claiming that the specific mediation is required there. The clone demonstrates a representation tie. Neither result validates universal BFG dynamics in nature.

11. A constructive synthetic witness and its matched direct control

After the preceding pilot, we designed a second, deliberately constructed generator with a fixed protocol and executable code [12]. Its favorable result is an existence example relative to a declared restricted rival class, not an independent preregistered test of a theory. Let $s_{1,t}$ and $s_{2,t}$ be independently driven signs of amplitude one. Define $q_t = 1 + 0.5s_{1,t}s_{2,t}$ and evolve distinct mediator and higher states:

$$m_{t+1} = 0.6m_t + 0.4q_t, \quad h_{t+1} = 0.4h_t + 0.6\gamma m_t, \quad m_0 = h_0 = 1. \quad (19)$$

For $\gamma = 1$, the convex updates keep m and the unperturbed h within $[0.5, 1.5]$, while both sources remain active. At step 80, an equal external perturbation resets h to zero in all arms. Setting $\gamma = 0$ for 30 steps removes the $m \rightarrow h$ generating edge, without changing sources or mediator. Returning γ to one at step 110 restores the higher state. The prespecified score is the mean h over the final ten steps of each comparison window. In 1,000 matched 160-step sequences, the smallest full-minus-removed score at steps 100–109 was 0.550554 and the smallest restored-minus-continuously-removed score at steps 120–129 was 0.563750. Both exceed the fixed threshold 0.49; all source gates passed. These inequalities also follow from positivity and the contraction factors, so the simulated repetitions illustrate rather than establish the result.

The same-time mixed contrast $q(+,+) - q(+,-) - q(-,+) + q(-,-) = 2$ excludes the specified additive or interaction-free controls under controlled source assignments. A circular shift with its shifted history fixed likewise has zero contrast. The unit impulse response from q_t to h_{t+k} is $(g_1, g_2, g_3, g_4) = (0, 0.24, 0.24, 0.1824)$, with $g_2g_4 - g_3^2 = -0.013824$. A one-pole direct dyad $h_{t+1} = \lambda h_t + c q_{t-1}$ has zero determinant, but lacks the full model's memory and is not the decisive direct comparison.

$$r_t = 0.6^t + 0.4 \sum_{k=0}^{t-1} 0.6^{t-1-k} q_k, \quad h_{t+1} = 0.4h_t + 0.6\gamma r_t. \quad (20)$$

The matched-memory direct source-history control (20) contains no explicitly named mediator state, but has the same effective memory and receives the same perturbation, removal, and restoration schedule. It reproduces every observed higher path in all three arms for 1,000 matched trajectories, with maximum absolute error 6.661×10^{-16} . Independent controlled assignments give the same four impulse values and a higher-state mixed contrast 0.48 at lag two. The control therefore passes the signatures and interventions that the constructed system passes. The physical mediation claim is not identified by these observations: r is mathematically the eliminated m . It is an equivalent representation, not a distinct causal explanation proved false. The originally published restricted-rival pass assertion is corrected here and retained in version history [12].

Table 4. Corrected constructive test and logical reach.

Gate	Result	Scope
Source and higher persistence	1,000/1,000 source gates; full steady h in [0.50023,1.49982].	Stipulated signed sources and convex update.
Removal and restoration	Minimum contrasts 0.55055 and 0.56375; threshold 0.49.	Causal necessity within this generator.
Direct source history	Paths and intervention signatures tie within 6.661×10^{-16} .	Same effective memory; mediation unidentifiable.
One-pole diagnostic	Zero determinant versus -0.013824 .	Insufficient memory; not decisive for mediation.

This toy supplies a coherent higher-state construction satisfying the declared operational synthetic gates. The specific-mediation exclusion conjunct in (11) is not established by the tied matched-memory direct control. Its equations were stipulated, not derived from historical Level 0 or executed through the canonical five-object successor operator. No natural carrier was tested. The earlier unfavorable direct-dyad result remains unchanged. A selective intervention on an independently measurable physical mediator, beyond the shared input and reported paths here, would be needed for physical identification.

12. Formal completion and outstanding claims

The strongest presently established synthesis has four parts: a universal recursive grammar (1)–(2); a canonical finite successor candidate (3)–(7); an exact internal novelty theorem on a specified stratum (8)–(10); and a conditional criterion for irreducible higher closure (11) with a workable prospective test architecture. The latter is a complete *definition and decision framework*, not an unconditional theorem that all BFG successors are emergent.

Two independent boundaries remain. First, the canonical quadratic neutral completion is explicit in the later BFG corpus, but it is not uniquely entailed by the earliest qualitative Level-0 statements. A Level-0 formation functional can supply a local positive Hessian and thus a local Gram pullback, while its full quartic expansion need not equal the exact quadratic completion at arbitrary displacement [4,5]. Second, the constructed witness supplies removal and restoration by design, but its matched-memory direct control ties and no natural carrier or unrestricted rival audit has been established. These are obligations for a chosen realization, not missing algebra in Theorem 1.

$$\text{Universal classification rule} \neq \text{universal existence of irreducible emergence.} \quad (21)$$

A new paper may therefore accurately state that BFG's reclosure and emergence conditions have been integrated formally. It should not state that the integration proves ubiquitous emergence, a unique physical

neutral mediator, or empirical universality. A negative or indeterminate carrier result must remain reportable under the same fixed rule.

13. Conclusion

The BFG generator and finite reclosure can be read as complementary layers of one research program. The finite operator constructs candidates, and the generator-level criteria classify their possible higher closure. The neutral-compatible inheritance–novelty identity is exact. A new constructive example satisfies operational source, higher-state, removal, and restoration gates, but a fair direct source-history representation ties on all observed signatures and interventions. An earlier pilot also favored a direct dyad on its chosen endpoint. Thus synthetic causal removal is demonstrable within stipulated dynamics, whereas specific physical mediation and the complete canonical BFG emergence claim remain open. Universal formal closure with a precise emergence criterion is integrated; universal emergent realization remains open.

Appendix A. Symbol register

Table 5. Main symbols.

Symbol	Meaning
D, Coh, N, Γ	Constitutive differentiation, coherence, neutral compatibility, regulated exclusion.
$Pers, Cl_BFG$	Non-vanishing bounded persistence and recursive closure.
$\Xi, U_BFG, \perp$	Finite five-object state, successor map, terminal outcome.
K_P, Y_P, χ_{ij}	Restricted inherited formation, neutral load, Schur factor.
S, R_+, C_+	Source closures, proposed constitutive relation, higher closure candidate.
Q, N^*, Δ_{gen}	Prespecified higher endpoint, mechanistic rival class, paired removal contrast.

Appendix B. Reproducibility and provenance

The analytical proofs in Sections 4 and 7 require no simulation. The pilot numbers in Section 10 are transcribed from the frozen report [9] and its accompanying executable Python file `bfg_relational_ablation_pilot.py`, seed 20260929. The new constructive generator and frozen results in Section 11 are reproduced by `bfg_constructive_emergence_witness.py` under `CONSTRUCTIVE_TEST_PROTOCOL.md` [12]. This constructed model was selected to exhibit the declared gates and does not constitute independent confirmation. The earlier circular-shift figures are transcribed from [1] and its published numerical supplement. Figure 1 is an explanatory diagram, not data. No natural empirical dataset was analyzed for this manuscript.

Appendix C. Level-0 formation to neutral tangent sector

The BFG source [13, Section 3.1] gives a real finite-dimensional formed branch $F_0(\delta) = \frac{1}{2} \langle \delta, K_0 \delta \rangle + \frac{1}{4} \|\delta\|^4$ with simple lowest eigenvalue $\lambda_0 < 0$ and normalized eigenvector e_0 . At $\delta^* = \pm \sqrt{-\lambda_0} e_0$ its Hessian is

$$H_-^* = D^2 F_0(\delta^*) = K_0 - \lambda_0 I - 2\lambda_0 e_0 e_0^\dagger > 0. \quad (22)$$

Along e_0 its eigenvalue is $-2\lambda_0$; on every perpendicular eigenmode e_j it is $\lambda_j - \lambda_0 > 0$. Define the exact quadratic tangent form $Q^*(x) = \frac{1}{2} \langle x, H_-^* x \rangle$. Taylor's theorem gives $F_0(\delta^* + x) - F_0(\delta^*) = Q^*(x) + O(\|x\|^3)$, so equality to the quartic parent energy at finite displacement is not claimed.

Let D be a perturbation of this branch, let L map a neutral perturbation η to the source perturbation carrier, and let $M > 0$ be the stated neutral-channel metric. The first metric H_-^* is computed from Level 0; M and the two-piece mediation role must be identified for the chosen carrier. The tangent neutral completion is

$$J_{\tan}(D, \eta) = \frac{1}{2} \langle D - L\eta, H^*(D - L\eta) \rangle + \frac{1}{2} \langle \eta, M\eta \rangle. \quad (23)$$

Write $H^{1/2}$ and $M^{1/2}$ for the positive square roots of H^* and M . Set $d = H^{1/2} D$, $v = M^{1/2} \eta$, and $A^\dagger = H^{1/2} L (M^{1/2})^{-1}$. These are invertible coordinate changes in the declared quadratic sector; therefore (23) is precisely the published canonical functional (5) in normalized coordinates. In particular:

$$Y = A^\dagger A = H^{1/2} L M^{-1} L^\dagger H^{1/2}; \quad \min J_{\tan} = \frac{1}{2} \langle D, H^{1/2} (I + Y)^{-1} H^{1/2} D \rangle. \quad (24)$$

For the scalar Level-0 branch $K_0 = -1$ and $\delta^* = 1$, $H^* = 2$. If $L = I$ and the neutral piece is the same formed branch ($M = 2$), then $d = \sqrt{2} D$, $A^\dagger = I$, $Y = I$, and $\min J_{\tan} = D^2/2 = d^2/4$. This agrees with the quadratic part of the exact quartic two-piece minimum. Indeed $f(z) := F_0(1+z) - F_0(1) = z^2 + z^3 + z^4/4$ and, for $D \geq 0$, the unique global minimum of $f(D-\eta) + f(\eta)$ is

$$2f(D/2) = D^2/2 + D^3/4 + D^4/32. \quad (25)$$

To verify global minimality, write $\eta = D/2 + t$ and $z = D/2$. Then $f(z+t) + f(z-t) - 2f(z) = t^2(2 + 6z + 3z^2) + t^4/2 > 0$ for $t \neq 0$ and $z \geq 0$. Equations (23)–(24) are exact throughout the *quadratic tangent model*; equation (25) records the finite-displacement remainder of the full Level-0 formation. Thus Sections 3.1 and 3.2 are mathematically compatible with an explicit tangent-sector and whitening convention. The role choice and any distinct neutral metric M are not uniquely derived from Section 3.1 alone.

Appendix D. Declarations

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